To compare the overall relative advantages of a 6 fan/6 gas generator system with an 8 fan/8 gas generator system, an existing 8 fan design, from Reference 1, was modified to reflect the current study guideline of a 25 percent structural weight saving through the use of advanced composite material technology. This aircraft was then compared with the basepoint 6 fan airplane developed for this study. Table 1 summarizes the major elements of the results of this comparison.

TABLE 4. SUMMARY COMPARISON OF 6 FAN AND 8 FAN AIRCRAFT

PARAMETER	6-FANS	8-FANS
CRUISE MODE	2 L / C FANS	2 L / C FAN + 2 TJ
PROPULSION SYS T / W	4.95	6.09
PROPULSION SYS WT	26,300 LB	19,600 LB
FAN TIME CONSTANT	0.34	0.26
CRUISE FUEL FLOW	12,500 LB / HR	15,000 LB / HR
FUEL REQUIRED	15,850 LB	19,000 LB
NOISE	98.3 PNdb	98.5 PNdb
TOGW	106,000 LB	105,000 LB
AIRFRAME COST	$4,823,500	$5,104,000
PROPULSION COST	$4,326,300	$4,690,000
TOTAL COST	$9,149,800	$9,794,000
D O C	1.0	1.066
DISPATCH RELIABILITY	1.0	0.97 - 0.995

The data of Table 4 indicate that, as expected, the 8 fan arrangement has an installed propulsion system T/W ratio of over 20 percent better than the 6 fan system. The six fan system has a higher total installed thrust requirement than the 8 fan system to meet the higher takeoff weight and failure criteria with fewer remaining propulsion units. Due to the need to use two gas generators in the turbojet mode to provide adequate cruise thrust, however, the 8 fan system consumes about 20 percent more fuel than the 6 fan system. Because of the requirements for packaging 8 fans versus 6 and